

Activity — Read the Working Instrument

60 minutes · in pairs · open notebooks · **every answer cites a section number.** *“It says somewhere” is not a citation.*

Names: Lagdamen, Lawrence M. **Team:** 1 — Alcantara · Sanchez / 2 — Lagdamen · Lopez

TEAM 1 — Alcantara · Sanchez

Notebook 02 — The Credential and the Job, Rev. 2. Your names are on its header.

TEAM 2 — Lagdamen · Lopez

Notebook 01 — Assessment Hubs and Assessment Deserts. Your names are on its header.

PART A · EVERYONE TOGETHER 10 MINUTES

The extraction notebook (00) — where does the human judgment live?

A1. Notebook 00 runs on one rule that splits the work between the script and the human. Write the rule, and name the FILE where the human’s column judgment is recorded.

“The script does the mechanical part. The human does the semantic part.”

File name: - pdfplumber

A2. The harvester pulls every number as n1, n2, n3 ... instead of naming them “graduates” or “assessed.” Why? One sentence.

So that the extracted data are labeled properly and according to their respective edition, to prevent the script from extracting the necessary data from the inappropriate edition.

A3. flag_anomalies marks a suspicious row. What does it deliberately NOT do to that row — and what is the sentence the notebook gives as the reason?

It flags those rows without fixing or dropping either of them. The notebook says that if a flagged row is dropped, it becomes a data problem.

PART B · YOUR OWN NOTEBOOK 30 MINUTES

Hunt the three disciplines — with cell-level evidence

Answer the six items for YOUR team’s notebook only. Evidence means a section number plus a quoted sentence or a printed number.

TEAM 1 — Notebook 02 (skip if you are Team 2)

B1. FEATURES — §8 contains a one-question test that decides every feature. Copy it verbatim.

B2. FEATURES — Name the two feature sets. For which outcome is the job-context set circular, and why?

B3. FEATURES — One feature is banned from Outcome 2 in writing. Which one, what did the Rev. 1 SHAP plot show, and what is the label’s definition that makes it a leak?

B4. BASELINES — §9 fits four models on the same features and weights. List them in order. What must LightGBM beat to earn its place — and if it fails, what does the notebook say the paper reports?

B5. PIPELINE — Calibration is fitted on one block and the reliability diagram is drawn on another. Which blocks? And how many times is the test block touched per model, and for what?

B6. PIPELINE — Write the temporal split (§7) in numbers. Why does the calendar do the splitting, not a random draw?

TEAM 2 — Notebook 01 (skip if you are Team 1)

B1. PIPELINE — The header names the full path a reviewer must be able to walk, from source table to claim. Copy the chain, arrows and all.

3 · Load the extraction workbook and validate its schema → 4 · Reconciliation — recompute every published total → 5 · Roster harmonization → 6 · Panel construction — and the window guard → 7 · Descriptives, and the finding that motivates the paper → 8 · Is it overdispersed? Poisson before negative binomial → 9 · The negative binomial mixed model → 10 · Sensitivity — the three checks the proposal promised → 11 · Baselines first → 12 · Rolling-origin evaluation → 13 · Hierarchical reconciliation → 14 · The CY 2026 nowcast → 15 · The CY 2027 forecast — the only forward number that earns the word → 16 · CY 2028–2030 — conditional projections, not forecasts → 17 · Concentration — location quotients, Herfindahl, Gini → 18 · Convergence — and why one test is not enough → 19 · The policy note — and the inversion audit → 20 · Close out — claim ledger, run log, and what did not get done

B2. BASELINES — §11 names two baselines. What are they, and WHY are they fitted before any model rather than after?

- Naive — next year equals this year.
- Historical mean — next year equals the mean of the training years.

They are fitted **before any model** so the benchmark is fixed in advance and cannot be chosen afterward just to make the model look better.

B3. PIPELINE — §12 explains why a random 80/20 split on this panel would be worthless. Write the reason in one sentence (yours or the notebook's).

A random 80/20 split is worthless because it can place later years in the training set and earlier years in the test set, so **the model learns the future in order to predict the past**, which gives misleadingly good results.

B4. BASELINES — What does MASE below 1.0 mean in plain words? And why does the notebook refuse MAPE for regional volumes?

MASE below 1.0 means the model performs better than the naive “**same as last year**” forecast.

The notebook refuses **MAPE** because regional assessment volumes differ greatly in size, so percentage errors can become extremely large for small regions and make comparisons misleading. Instead, MASE is scale-free and compares the model's error directly with the naive forecast's error on the same series.

B5. REGISTERS — Which of the five registers does a baseline forecast belong to? From §11, copy one “You may write” sentence and one “You may not write” sentence.

A baseline forecast belongs to the **FORECAST** register, specifically **FORECAST (baseline)**.

You may write:

“Against a naive baseline, model M achieves MASE of ...”

You may not write:

“any accuracy claim without the baseline beside it.”

B6. PIPELINE — The window guard in §6: which years may PANEL_COV never contain, and how often is that assertion checked?

PANEL_COV must **never contain any year before 2017**. The assertion is checked **every time the panel is built**, not just once, so an invalid year cannot silently enter the modeling dataset later

PART C · CROSS-BRIEF 15 MINUTES

Teach the other team one discipline

Pick ONE discipline you found (features, pipeline, or baselines). Teach it to the other team in two minutes: evidence first — section number, quoted sentence or printed number — then the lesson in one sentence.

C1. The discipline WE taught, and the evidence we used:

Evidence — §11: The notebook uses two pre-declared baselines: **naive** (“**next year equals this year**”) and **historical mean** (“**next year equals the mean of the training years**”). It states that they are fitted **before any model** so nobody can choose a baseline afterward just to make the model look better. BDAT05_S6_Activity_Sheet

Lesson: A predictive model only earns its value if it beats a benchmark that was fixed **before** modeling, not selected afterward.

C2. The discipline WE WERE taught, and one specific place it will apply in OUR OWN track project (P4–P7):

Hand in: this sheet, both names on it, before wrap-up. Carry answer C2 into your next milestone — I will look for it there.